WORKED SOLUTION

# STEP I 1998, Question 1

Pure • Counting Digit Patterns

> **STEP I 1998, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/98-s1-q1

## The key idea

Don't try to list numbers. Instead, first choose *which* two digits appear, then count the five-digit strings that use both of them. The only snag is that a number can't start with 0, so we split the count by whether 0 is one of the two digits. And don't forget to check both ends of the range.

## Solution

Every integer from 10 000 to 99 999 has exactly five digits, with a non-zero first digit. The one remaining integer in the range is 100 000, which we deal with at the end.

A five-digit number contains exactly two different digits when its digits all come from some pair $\{a, b\}$ and both $a$ and $b$ actually appear. Each such number determines its pair, so we can count pair by pair and add.

### Case 1: neither digit is 0

There are $\binom{9}{2} = 36$ ways to choose two different digits from $1, 2, \ldots, 9$. For a fixed pair $\{a, b\}$, each of the five positions can be $a$ or $b$, giving $2^5 = 32$ strings. Two of these use only one digit ($aaaaa$ and $bbbbb$), so $32 - 2 = 30$ use both. None of them starts with 0, so all 30 are genuine five-digit numbers. This case gives

$$36 \times 30 = 1080.$$

**?** **Ask yourself** Why can't we treat the pair $\{0, a\}$ in exactly the same way?

Because half of the 32 strings made from 0 and $a$ start with 0, and those aren't five-digit numbers (for example, $0a0aa$ is really a four-digit number). So we have to fix the first digit before counting.

#### Case 2: one of the digits is 0

The other digit $a$ can be any of $1, \ldots, 9$: 9 choices. The first digit can't be 0, so it must be $a$. Each of the remaining four positions can be 0 or $a$, giving $2^4 = 16$ possibilities. Exactly one of these, $aaaa$, contains no 0, which would leave only one distinct digit. So there are $16 - 1 = 15$ numbers for each $a$, and this case gives

$$9 \times 15 = 135.$$

#### The endpoint 100 000

The range is inclusive, and 100 000 uses just the digits 1 and 0, so it counts too.

#### Total

$$1080 + 135 + 1 = 1216.$$

So there are **1216** such integers.

## Checks

- A computer check that runs through every integer from 10 000 to 100 000 and counts those whose digits form a set of size 2 gives 1216. It also confirms the split: 1080 without a 0, 135 with one, plus 100 000.
- Try the same method on a smaller case you can list by hand. For two-digit numbers it gives $\binom{9}{2}(2^2 - 2) + 9(2^1 - 1) = 72 + 9 = 81$. That matches the direct count: 90 two-digit numbers, minus the 9 repdigits $11, 22, \ldots, 99$.

## Common slips

- Forgetting 100 000, giving 1215. Always look at the endpoints of an "inclusive" range.
- Counting $\binom{10}{2} \times 30 = 1350$, which includes numbers that start with 0.
- Subtracting 2 instead of 1 in Case 2: the all-zero string was already ruled out when we fixed the first digit as $a$.

## Beyond STEP

The trick of subtracting the one-digit strings from all strings over $\{a, b\}$ is the simplest case of the inclusion–exclusion principle. In general, the number of strings of length $n$ that use *all* of $k$ given symbols is $\sum_{j=0}^{k} (-1)^j \binom{k}{j} (k - j)^n$, which is $k!$ times a Stirling number of the second kind. These counts appear throughout combinatorics and probability, for instance in the coupon collector problem. For more of this kind of

counting, try Grimmett and Welsh, *Probability: An Introduction*, or the combinatorics questions in Siklos, *Advanced Problems in Mathematics*.

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